

From Six-Parameter Somos–4 Data to Weierstrass, Jacobi, Ramanujan Theta Representations

A Constructive Workflow

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Abstract

A generic Somos–4 orbit with four freely prescribed initial values satisfies

$$H_{n+4}H_n = \alpha H_{n+3}H_{n+1} + \tau H_{n+2}^2.$$

An earlier workflow treated the normalized case $\alpha = 1$ and organized the passage from a direct algebraic generating function to a genus-one curve, a Weierstrass sigma representation, and a Jacobi θ_1 representation. The present paper develops the corresponding six-parameter workflow and adds a further endpoint suggested by Michael Somos: Ramanujan’s general theta function

$$f_R(a, b) = \sum_{k \in \mathbb{Z}} a^{k(k+1)/2} b^{k(k-1)/2}, \quad |ab| < 1.$$

Starting from $(\lambda, m, r, \eta, \alpha, \tau)$, we separate three layers that must not be conflated: the Somos orbit, its elliptic curve, and a chosen power-series lift. A weighted transport from the five-parameter algebraic generating function introduces a sixth root of α in that particular lift, whereas the recurrence and a birational quartic model remain rational in the six parameters. The elliptic part of the workflow then proceeds through periods, a translation point, Abel–Jacobi coordinates, and the classical sigma formula. The Jacobi endpoint is converted exactly into Ramanujan form, with moving arguments a_n, b_n satisfying the fixed-product relation $a_n b_n = q^2$, where $q = e^{\pi i \tau_{\text{mod}}}$. This exposes both a bilateral q -series and an infinite-product representation of the same Somos–4 orbit. The emphasis is on normalization contracts, branch and indexing checks, and a reproducible computational plan. PARI/GP routines and numerical tables are left as explicit placeholders for a companion implementation.

1 Introduction

The sigma-function description of generic Somos–4 sequences is classical. The practical task, however, is not merely to quote the existence of such a description. Starting from concrete recurrence data, one must identify the correct invariant curve, choose compatible coordinates and periods, locate the translation point, fix Abel–Jacobi lifts, and recover the elementary gauge factors. Each step carries a convention whose unnoticed change may preserve some low-order checks while invalidating the final formula.

The preceding workflow paper carried out this program for the normalized recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2,$$

starting from the direct algebraic generating function with prescribed $(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$. Two developments motivate the present follow-up. First, the generating-function construction has been extended to the six-parameter recurrence with a free coefficient α .

Second, Michael Somos pointed out that Ramanujan theta functions provide a natural and less frequently documented endpoint of the workflow. The latter is not only a change of notation: Ramanujan's $f_R(a, b)$ simultaneously displays the bilateral series and Jacobi triple product, while the sequence index enters explicitly through its two arguments.

The resulting workflow is

$$(\lambda, m, r, \eta, \alpha, \tau) \mapsto (J, \mathcal{C}) \mapsto E \mapsto P \mapsto (\omega_1, \omega_2; z_0, \delta) \mapsto \begin{cases} \sigma, \\ \theta_1, \\ f_R(a, b). \end{cases}$$

The six-parameter entry requires care. The direct transported generating series lives naturally over a radical extension $q_6^6 = \alpha$, but the ordinary Hankel orbit and a rational quartic model live over

$$K = \mathbb{Q}(\lambda, m, r, \eta, \alpha, \tau).$$

Accordingly, this paper distinguishes statements about a generating series from statements about the curve or recurrence.

1.1 Scope and contribution

This paper provides:

- a workflow-level synthesis of the six-parameter generating-function update with the algebraic-to-sigma construction;
- a rational quartic entry point for the six-parameter orbit, together with explicit check-points separating it from a radical power-series lift;
- the exact conversion from the Jacobi θ_1 endpoint to Ramanujan's general theta function $f_R(a, b)$;
- the fixed-product law $a_n b_n = q^2$ and the associated bilateral-series and infinite-product forms;
- a verification architecture for recurrence, curve, sigma, Jacobi, and Ramanujan evaluations; and
- marked placeholders for the later PARI/GP companion and numerical examples.

It does not claim a new proof of the classical sigma representation, nor does it assert that every six-parameter algebraic lift requires radicals. The non-descent statement concerns the particular weighted transport used below.

2 Six-parameter input and the three mathematical layers

Let

$$H_{n+4}H_n = \alpha H_{n+3}H_{n+1} + \tau H_{n+2}^2, \quad (H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta). \quad (1)$$

We assume throughout the main workflow that all displayed denominators are nonzero and that the resulting genus-one curve is nonsingular. Degenerate cases require separate projective or limiting treatments.

Three related objects must be kept distinct:

ordinary Hankel orbit, genus-one curve, chosen generating series.

The recurrence generates $H_n \in K$. The six-parameter follow-up also gives a quartic model over K . By contrast, the weighted algebraic-series transport uses q_6 with $q_6^6 = \alpha$ and generically has coefficients outside K . This distinction is essential for a correct workflow.

2.1 Weighted generating-series transport

Let $G_{1;\lambda,m,r,\eta,\tau}(x)$ denote the five-parameter series. Put

$$\lambda' = \lambda, \quad m' = q_6^{-2}m, \quad r' = q_6^{-6}r, \quad \eta' = q_6^{-12}\eta, \quad \tau' = q_6^{-8}\tau, \quad q_6^6 = \alpha, \quad (2)$$

and define

$$G_{\alpha;\lambda,m,r,\eta,\tau}(x) = G_{1;\lambda',m',r',\eta',\tau'}(q_6x). \quad (3)$$

If $b_k = q_6^k a_k$, row and column factorization gives

$$H_N(b) = q_6^{N(N-1)} H_N(a), \quad D_N(b) = q_6^{N^2} D_N(a). \quad (4)$$

Consequently, the first four ordinary determinants remain (λ, m, r, η) and satisfy equation (1). The inherited companion law is graded:

$$D_N(b) = q_6^{-3N-2} H_{N+2}(b). \quad (5)$$

Already $b_1 = q_6^{-5}r$, so this transported branch does not generically descend to K . Nothing in this observation prevents the orbit and curve from descending.

3 Reduced dynamics and rational curve model

Define the gauge-invariant variables

$$x_n = \frac{H_{n-1}H_{n+1}}{H_n^2}. \quad (6)$$

Equation (1) implies

$$x_{n+2}x_nx_{n+1}^2 = \alpha x_{n+1} + \tau, \quad x_{n+2} = \frac{\alpha x_{n+1} + \tau}{x_nx_{n+1}^2}. \quad (7)$$

The natural first integral is

$$J = x_nx_{n+1} + \alpha \left(\frac{1}{x_n} + \frac{1}{x_{n+1}} \right) + \frac{\tau}{x_nx_{n+1}}. \quad (8)$$

For the initial determinants,

$$x_2 = \frac{\lambda r}{m^2}, \quad x_3 = \frac{m\eta}{r^2}, \quad (9)$$

so that

$$J = \frac{\lambda\eta}{mr} + \alpha \frac{m^2}{\lambda r} + \alpha \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (10)$$

A rational quartic model compatible with the six-parameter follow-up is

$$\mathcal{C}_{J,\alpha,\tau} : \quad y^2 = (X^2 - JX + \tau)^2 - 4\alpha^2 X. \quad (11)$$

For $\alpha = 1$, this reduces to the quartic used in the preceding workflow.

TODO: In the code session, verify equations (8) and (11) symbolically from consecutive QRT states and record the exact state-to-quartic map. The normalization must be checked rather than inferred solely by specialization.

4 The constructive workflow

4.1 Workflow Step 1: Validate the six-parameter input

Read $(\lambda, m, r, \eta, \alpha, \tau)$, generate several exact recurrence values, and test every denominator required by the selected affine charts. Compute x_2, x_3 and J from equations (9) and (10).

Checkpoint. Verify equation (8) for several consecutive pairs generated by equation (7). Verify independently that the H_n and x_n routes agree.

4.2 Workflow Step 2: Select the orbit, curve, or series route

If only the elliptic or transcendental representation is required, work over K with equation (11). If the direct algebraic generating function is also required, construct equation (3) over $K(q_6)$ and retain the graded companion law equation (5). Do not use the literal five-parameter companion identity after transport.

Checkpoint. Confirm $(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$ and

$$H_5 = \frac{\alpha\eta m + \tau r^2}{\lambda}, \quad H_6 = \frac{\alpha H_5 r + \tau \eta^2}{m}.$$

For the transported series, confirm $D_N = q_6^{-3N-2} H_{N+2}$ at the first orders.

4.3 Workflow Step 3: Convert the rational quartic to Weierstrass form

Choose one point at infinity as origin and compute a birational map

$$(X, y) \longleftrightarrow (u, v), \quad E : v^2 = 4u^3 - g_2 u - g_3. \quad (12)$$

The formulas must specialize to those of the normalized workflow at $\alpha = 1$.

TODO: Insert the exact six-parameter birational maps, $g_2(J, \alpha, \tau)$, $g_3(J, \alpha, \tau)$, the elliptic discriminant, and the affine exceptional set after symbolic derivation and PARI/GP verification.

Checkpoint. Verify both map compositions modulo the defining curve equations. Check that the quartic discriminant and $g_2^3 - 27g_3^2$ agree up to the stated scaling, and verify the j -invariant independently.

4.4 Workflow Step 4: Identify the fixed elliptic translation

Map the QRT state (x_n, x_{n+1}) to a point $Q_n \in E$. Determine the fixed point $P \in E$ satisfying

$$Q_{n+1} = Q_n + P. \quad (13)$$

The sign convention for the second quartic coordinate fixes whether a step is represented by P or $-P$.

TODO: Derive and insert the explicit six-parameter state map and translation point. Verify that the formulas reduce to $P = ((J^2 - 4\tau)/12, 1)$ when $\alpha = 1$ in the text convention.

Checkpoint. For several exact QRT iterates, verify that Q_n lies on E and that $Q_{n+1} - Q_n = P$ is independent of n .

4.5 Workflow Step 5: Compute and normalize the period lattice

Compute an oriented basis of full periods $(2\omega_1, 2\omega_2)$ for equation (12) and set

$$\tau_{\text{mod}} = \frac{\omega_2}{\omega_1}, \quad \text{Im}(\tau_{\text{mod}}) > 0. \quad (14)$$

Only the lattice is intrinsic; the ordered basis is a retained workflow choice. Every later sigma, theta, and Ramanujan evaluation must use the same basis.

Checkpoint. Reconstruct g_2, g_3 from the lattice and compare with the algebraic values. State explicitly whether software returns full or half periods.

4.6 Workflow Step 6: Compute Abel–Jacobi coordinates

Choose lifts z_2 and δ such that

$$(\wp(z_2), \wp'(z_2)) = Q_2, \quad (\wp(\delta), \wp'(\delta)) = P,$$

and define, for the present H_1 -based indexing,

$$z_0 = z_2 - 3\delta \pmod{\Lambda}. \quad (15)$$

Then Q_n is represented by $z_0 + (n+1)\delta$.

Checkpoint. Compare both Weierstrass coordinates, not only the first coordinate, for several n . This fixes the sign of δ .

4.7 Workflow Step 7: Fix the sigma normalization

Use the classical representation

$$H_n = AB^n \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}. \quad (16)$$

Put

$$S_n = \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}.$$

When the displayed factors are nonzero, the first two initial values give

$$B = \frac{mS_1}{\lambda S_2}, \quad A = \frac{\lambda^2 S_2}{mS_1^2}. \quad (17)$$

The values $H_3 = r$ and $H_4 = \eta$ are then independent consistency checks. The coefficient α is encoded in the elliptic data and the translation step, not in an extra arbitrary sigma gauge.

Checkpoint. Compare equation (16) with exact recurrence values and verify

$$x_n = \wp(\delta) - \wp(z_0 + n\delta) \quad (18)$$

under the selected normalization. If the six-parameter scaling changes this identity, record the corrected factor explicitly rather than absorbing it silently.

4.8 Workflow Step 8: Convert sigma to Jacobi θ_1

For the chosen ordered half periods, let

$$\eta_1 = \zeta(\omega_1), \quad c = \frac{2\omega_1}{\pi\theta_1'(0 \mid \tau_{\text{mod}})}.$$

The normalization constant is

$$\sigma(z) = \exp\left(\frac{\eta_1 z^2}{2\omega_1}\right) c \theta_1\left(\frac{\pi z}{2\omega_1} \mid \tau_{\text{mod}}\right). \quad (19)$$

Define

$$A_0 = \frac{\pi\delta}{2\omega_1}, \quad B_0 = \frac{\pi z_0}{2\omega_1}. \quad (20)$$

Then

$$H_n = K_0 q_0^{n^2} r_0^n \theta_1(A_0 n + B_0 \mid \tau_{\text{mod}}), \quad (21)$$

where

$$K_0 = A c \exp\left(\frac{\eta_1 z_0^2}{2\omega_1}\right), \quad (22)$$

$$q_0 = \exp\left(\frac{\eta_1 \delta^2}{2\omega_1}\right) \sigma(\delta)^{-1} = (c \theta_1(A_0 \mid \tau_{\text{mod}}))^{-1}, \quad (23)$$

$$r_0 = B \exp\left(\frac{\eta_1 z_0 \delta}{\omega_1}\right). \quad (24)$$

Checkpoint. Evaluate the sigma and Jacobi forms through independent code paths. Libraries that use the nome rather than τ_{mod} , or a rescaled first argument, require an explicit adapter.

4.9 Workflow Step 9: Convert Jacobi θ_1 to Ramanujan $f_{\text{R}}(a, b)$

Set

$$q = e^{\pi i \tau_{\text{mod}}}, \quad |q| < 1, \quad (25)$$

and define Ramanujan's general theta function by

$$f_{\text{R}}(a, b) = \sum_{k=-\infty}^{\infty} a^{k(k+1)/2} b^{k(k-1)/2}, \quad |ab| < 1. \quad (26)$$

Its Jacobi triple product is

$$f_{\text{R}}(a, b) = (-a; ab)_{\infty} (-b; ab)_{\infty} (ab; ab)_{\infty}. \quad (27)$$

For the convention in equation (19), the exact bridge is

$$\theta_1(z \mid \tau_{\text{mod}}) = i q^{1/4} e^{-iz} f_{\text{R}}(-e^{2iz}, -q^2 e^{-2iz}). \quad (28)$$

Put $z_n = A_0 n + B_0$ and define

$$a_n = -e^{2iz_n}, \quad b_n = -q^2 e^{-2iz_n}, \quad (29)$$

$$K_R = i K_0 q^{1/4} e^{-iB_0}, \quad r_R = r_0 e^{-iA_0}. \quad (30)$$

Substitution into equation (21) gives the Ramanujan endpoint

$$\boxed{H_n = K_R q_0^{n^2} r_R^n f_R(a_n, b_n).} \quad (31)$$

The moving arguments satisfy the fixed-product law

$$\boxed{a_n b_n = q^2}, \quad (32)$$

which is independent of the Somos index n .

Checkpoint. Verify equation (28) first for generic complex test arguments, then compare equation (31) with the Jacobi, sigma, and recurrence values. Test convergence as the bilateral truncation grows.

5 What the Ramanujan endpoint adds

The conversion is exact, but it reveals structure that is less visible in the sigma notation. From equations (26) and (30),

$$f_R(a_n, b_n) = \sum_{k \in \mathbb{Z}} (-1)^k q^{k(k-1)} e^{2ik(A_0 n + B_0)}. \quad (33)$$

Thus

$$H_n = K_R q_0^{n^2} r_R^n \sum_{k \in \mathbb{Z}} (-1)^k q^{k(k-1)} e^{2ikB_0} e^{2ikA_0 n}. \quad (34)$$

The sequence index occurs explicitly in the exponential factor $e^{2ikA_0 n}$. It is therefore natural to regard equation (34) as a bilateral theta or Fourier expansion in the discrete index, without assigning it an interpretation beyond the exact identity.

The product side is equally direct. Since $a_n b_n = q^2$,

$$f_R(a_n, b_n) = (e^{2iz_n}; q^2)_\infty (q^2 e^{-2iz_n}; q^2)_\infty (q^2; q^2)_\infty. \quad (35)$$

Hence the same Somos orbit admits Weierstrass, Jacobi, bilateral q -series, and infinite-product descriptions.

5.1 Special Ramanujan functions

Ramanujan's classical specializations include

$$\phi(q) = f_R(q, q), \quad \psi(q) = f_R(q, q^3), \quad f(-q) = f_R(-q, -q^2).$$

The generic arguments in equation (30) do not usually reduce to these special forms. Such reductions require additional arithmetic relations among q , A_0 , and B_0 . They should therefore be treated as special modular cases rather than as part of the generic workflow.

TODO: Investigate, after the generic implementation is verified, whether the classical Somos-4 curve or other arithmetic examples yield useful ϕ , ψ , eta-product, or modular-equation specializations. Do not claim such a reduction without an exact parameter relation.

6 Algorithm summary

- Step 1.** Read $(\lambda, m, r, \eta, \alpha, \tau)$ and generate exact recurrence data.
- Step 2.** Compute x_2, x_3 and the invariant J .
- Step 3.** Decide whether the calculation needs the radical generating-series lift or only the rational orbit and curve.
- Step 4.** Construct the rational quartic and test nonsingularity.
- Step 5.** Compute and verify a birational Weierstrass model.
- Step 6.** Identify the fixed translation point from the QRT dynamics.
- Step 7.** Compute and orient the period lattice.
- Step 8.** Compute z_0 and δ , retaining signs and period representatives.
- Step 9.** Determine A, B and verify all four initial values.
- Step 10.** Convert sigma to Jacobi form using one stated normalization contract.
- Step 11.** Convert Jacobi form to Ramanujan $f_R(a, b)$ and verify $a_n b_n = q^2$.
- Step 12.** Compare recurrence, sigma, Jacobi, Ramanujan-series, and Ramanujan-product evaluations.

7 Computational architecture

The companion implementation should keep exact and numerical stages separate. The proposed components are:

- exact Somos recurrence and QRT iteration for (α, τ) ;
- exact invariant and quartic construction;
- exact forward and inverse birational maps;
- exact curve discriminant and j -invariant checks;
- numerical periods and elliptic logarithms at controlled precision;
- sigma evaluation using the selected lattice;
- an independent Jacobi θ_1 implementation or adapter;
- direct bilateral summation of $f_R(a, b)$;
- optional evaluation by the triple product; and
- a consolidated error report against exact recurrence values.

7.1 PARI/GP placeholders

```
\\ TODO: six-parameter recurrence and invariant
somos6_step(v, alpha, tau) = { /* placeholder */ };
qrt6_step(x, y, alpha, tau) = { /* placeholder */ };
qrt6_invariant(x, y, alpha, tau) = { /* placeholder */ };

\\ TODO: rational quartic and verified Weierstrass maps
quartic6_poly(J, alpha, tau, X) =
  (X^2 - J*X + tau)^2 - 4*alpha^2*X;
quartic6_to_weierstrass(X, Y, J, alpha, tau) =
  { /* placeholder */ };
weierstrass_to_quartic6(u, v, J, alpha, tau) =
  { /* placeholder */ };

\\ Ramanujan's general theta function by bilateral truncation
ramanujan_f(a, b, N = 40) =
{
  sum(k = -N, N,
    a^(k*(k+1)/2) * b^(k*(k-1)/2));
};

\\ theta_1 via the exact Ramanujan conversion used in the text
ramanujan_theta1(tau_mod, z, N = 40) =
{
  my(q = exp(Pi*I*tau_mod));
  I*q^(1/4)*exp(-I*z)
  * ramanujan_f(-exp(2*I*z),
    -q^2*exp(-2*I*z), N);
};

\\ TODO: compare recurrence, sigma, Jacobi, Ramanujan sum and product
workflow6_demo(lambda, m, r, eta, alpha, tau, N = 10, prec = 80) =
  { /* placeholder */ };
```

TODO: In the dedicated code session, replace every placeholder, add adaptive bilateral truncation and product truncation, and report numerical errors as a function of precision and truncation order.

8 Numerical examples

8.1 Normalized classical Somos-4

For

$$(\lambda, m, r, \eta, \alpha, \tau) = (1, 1, 1, 1, 1, 1),$$

the workflow must reproduce

$$1, 1, 1, 1, 2, 3, 7, 23, 59, \dots$$

This example also provides a regression test against the earlier workflow.

TODO: Insert the recomputed periods, Abel–Jacobi data, sigma constants, Jacobi constants, nome q , several (a_n, b_n) pairs, and a convergence table for the Ramanujan sum and product. Recompute rather than copying rounded values from the earlier manuscript.

8.2 A genuinely six-parameter example

Choose rational nonzero input with $\alpha \neq 1$ for which the quartic is nonsingular and the first several recurrence denominators do not vanish. The example should show separately:

1. exact recurrence and QRT checks over K ;
2. the rational quartic and Weierstrass invariants;
3. the radical coefficient $b_1 = q_6^{-5}r$ of the transported series;
4. numerical sigma and Jacobi evaluations; and
5. independent Ramanujan-series and product evaluations.

TODO: Select and verify the example during the PARI/GP session.

9 Reliability and reproducibility

The principal failure modes are now broader than in the normalized workflow:

- confusing the sixth root q_6 used in the generating-series transport with the elliptic nome $q = e^{\pi i \tau_{\text{mod}}}$;
- applying the ungraded companion identity after weighted transport;
- silently changing the quartic or Weierstrass scaling when α is introduced;
- reversing the QRT translation by changing the sign of a second coordinate;
- mixing full and half periods;
- using inconsistent Abel–Jacobi representatives;
- confusing q with q^2 in the Ramanujan arguments; and
- comparing two formulas that were derived through the same code path and therefore do not constitute independent verification.

The notation q_6 is reserved for $q_6^6 = \alpha$, while q denotes the nome. This typographical separation should be maintained in both manuscript and code.

10 Discussion and future directions

The six-parameter update sharpens the separation between algebraic and transcendental data. The ordinary Hankel orbit and its rational quartic may be handled over K , while a distinguished generating-series lift can retain a radical orientation invisible to ordinary Hankel determinants. Once the curve and translation are fixed, the sigma and Jacobi stages proceed as in the normalized case, but with modified elliptic invariants.

The Ramanujan endpoint adds a complementary view. It packages the same theta value as a bilateral series and an infinite product, with a constant nome product $a_n b_n = q^2$. This may be particularly useful when the elliptic data have arithmetic or modular specializations. The generic identity should be established and numerically checked first; special identities involving ϕ , ψ , $f(-q)$, eta products, or modular equations belong to a later arithmetic analysis.

11 Conclusion

The extended workflow begins with six freely varying parameters, preserves the rational Somos orbit and elliptic curve, and tracks explicitly where a radical enters the transported algebraic generating series. It then carries the orbit through QRT dynamics, a Weierstrass model, periods, Abel–Jacobi coordinates, and the classical sigma representation. The final conversion

$$\sigma \longleftrightarrow \theta_1 \longleftrightarrow f_R(a, b)$$

is exact under one stated normalization contract. In Ramanujan form, the sequence is

$$H_n = K_R q_0^{n^2} r_R^n f_R(a_n, b_n), \quad a_n b_n = q^2,$$

which exposes both the bilateral q -series and infinite-product structure. The remaining work is deliberately concrete: derive and verify the explicit six-parameter Weierstrass map and translation point, implement independent PARI/GP evaluation paths, and replace the marked placeholders with checked numerical data.

Statement on AI assistance and computational tools

This draft was prepared with AI-assisted synthesis, L^AT_EX editing, and normalization checking. The author remains responsible for the mathematical statements, references, computational verification, and final text. The planned companion implementation will use PARI/GP for exact algebra and high-precision elliptic computations.

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